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Paleomagnetic investigations of the tertiary and quaternary igneous rocks: IV. The tertiary volcanics of the Lausitz area, Germany [1])

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With 8 figures and 2 tables

Zusammenfassung

Die untersuchten Basalte und Phonolite der Lausitz gehören zu den Ausläufern eines starken tertiären Vulkanismus mit dem Zentrum im Böhmischen Mittelgebirge. An Proben von 30 verschiedenen Lokalitäten werden paläomagnetische sowie petrologische und erzmikroskopische Untersuchungen durchgeführt. Daraus werden Magnetisierungseinrichtungen erhalten, die auf engstem Raum inversen und normalen Charakter nebeneinander aufweisen. Im gleichen Maße wechseln paläomagnetisch stabile und instabile Proben miteinander. Die Ergebnisse der petrologischen und erzmikroskopischen Untersu-

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chungen führen in vielen Fällen zur Klärung der Ursachen für dieses unterschiedliche Stabilitätsverhalten und geben außerdem erste Hinweise über die stattgefundenen Inversionsmechanismen.

Abstract

The basalts and phonolites of the Lausitz area belong to the Bohemian volcanic province of which the centre of activity lay in the České Středohoří. Samples from 30 localities were investigated, petrological, ore microscopical and palaeomagnetic studies being carried out. Normal and reversed directions were found. Magnetically stable and unstable samples occur. In many cases the petrological and ore microscopy studies can lead to an explanation of these stability variations. They also have a direct bearing on the origin of the reversed and normal directions found.

Résumé

Les basaltes et phonolites de Lausitz appartiennent à la province volcanique de la Bohême du nord, dont l'ancien centre d'activité se trouve dans les České Středohoří. Sur les échantillons prélevés à trente affleurements les études pétrologiques, minéralogiques et paléomagnétiques ont été faites. Des directions d'aimantation normales et renversées ont été trouvées. Le même affleurement montre les échantillons stabiles et instabiles. Les études pétrologiques donnent des explications. Les mêmes études s'appliquent aux problèmes d'origine des directions normales et renversées.

Краткое содержание

Базальты и фонолиты в районе Лаузица залегают в зоне активного третичного вулканизма, центр которого располагался в средних горах Богемии. Были исследованы 30 проб петрографически, а также проведены палеомагнитные измерения на них. Т. о. были установлены направления намагниченности. Во многих случаях результаты петрологического и минераграфического исследований подтверждают результаты палеомагнитных измерений.

Introduction

The Tertiary volcanic rocks of the Lausitz area in the eastern part of Germany belong to the north Bohemian province. If the outcrops are restricted in number and areal extent their near central position makes them an integral part of any study of the Bohemian province as a whole. Currently investigation of the more extensive developments in Poland and Czechoslovakia is in progress. Palaeomagnetic measurements on some of the Lausitz phonolites and basalts have already been carried out by Vollstädt (1965, 1967) as part of a study of palaeomagnetic stability. One result of this work was a preliminary virtual pole position (Vollstädt, 1966). The relation of the Lausitz volcanics to the Tertiary vulcanicity of central Europe is shown in fig. 1.

The Lausitz volcanics comprise tuffs, basalts, both intrusive and extrusive, and phonolites. Although the tuffaceous rocks because of their friable character are but poorly exposed, the basalts and phonolites, necks

and plugs in particular, tend to form topographic prominences often densely wooded. There are numerous quarries, both working and abandoned, which with their access roads provide a means of collecting unweathered, if not necessarily unaltered, material. The samples on which this account is based was collected during the Easter of 1965. The magnetic measurements were carried out in Newcastle and the petrographical studies in Potsdam.

Outline of geology

Across central Europe in the vicinity of the Variscan folds, from the Eifel in Western Germany to Poland in the east lies a wide zone in which during Tertiary times there was a considerable development of volcanic activity (fig. 1 a). This eruptive activity was at its maximum during the time near the end of the Oligocene and into early Miocene. The volcanics of the Lausitz area, developed particularly in the southeastern part of the region, form but a small part of this activity and may best be regarded as part of the volcanic province of north Bohemia. This province extends roughly from the Doupovské hoŕý (Duppauer Gebirge) in the west to near Opole in Poland in the east. Further to the west following the same tectonic trend there are the volcanics of the Kaiserstuhl and Hegau which are of a petrologically similar type, although younger in age.

The maximum volcanic development in northern Bohemia is in the Doupovské hoŕý and in the České Středohoří (Mittelgebirge), and it is in these regions where most is known about their stratigraphical relationships. There, three phases of volcanic activity have been recognized and related in a general way to the Savian, Attic and Rhodanian (or even Vlašskian) tectonic phases. The first volcanic phase is the most extensive in time and space and in the variety of petrological types developed. In northern Bohemia it is now considered to cover the time range lower Aquitanian to Burdigalian (KOPECKÝ 1966), although earlier opinion considered the time extent somewhat greater. In this respect it is important to notice that the few radiometric age determinations from Poland (URRY 1936) indicate volcanic activity of late Oligocene age.

The second phase volcanics are more restricted in extent and type. They range in age from top Miocene into the lower Pliocene, but where direct stratigraphical control is lacking they cannot be distinguished petrographically from rocks of the first phase. The volcanics, of both phases, suffered strong erosion and in consequence only relicts of the more resistant lavas and isolated necks are found in Bohemia. This is more particularly true of the České Středohoří region with which the Lausitz area is most closely linked. Occasionally, xenoliths of Neogene age are found in neck breccias, and in one instance, carbonized wood fragments (Glyptostrobus europeus) help to fix the age of the volcanics.

In the third and final volcanic phase found in the peripheral regions, the Cheb basin and Nižký Jeseníky, extrusives were general. Almost

unaltered volcanic products lie upon Terrace gravels dated as upper Pliocene or Quaternary. Activity closed before the last Würm interstadial.

In the Lausitz area itself there is little exact age information to help fix the age of the effusives. The volcanic sequence began with the development of tuff which is but seldom exposed. Near Seifhennersdorf

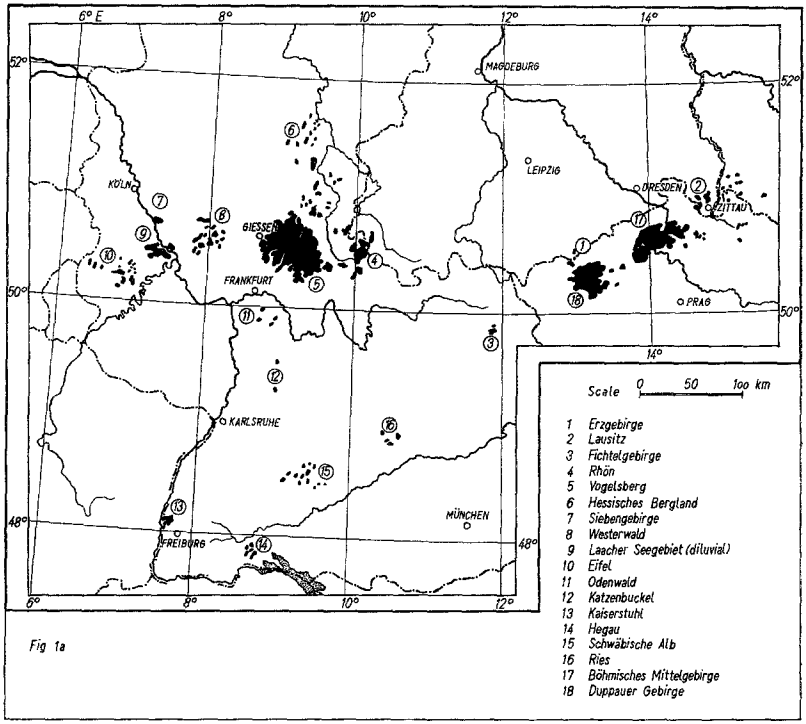


Fig 1a

Fig. 1 a. Sketch map indicating the Tertiary volcanic centres of central Europe.

however, basaltic tuff is interbedded in the upper Oligocene Polierschiefer whose age is given as Chattian on palynological grounds (PIETSCH 1962). Above follows a sequence of basaltic flows so that the vulcanicity lies in the time range from upper Chattian to lower Aquitanian. Nevertheless, as in the České Středohoří (Böhm. Mittelgebirge) and Doupovské Hoří (Duppauer Gebirge) later eruptions cannot be ruled out at the present state of knowledge. The volcanic episode appears to have been brought to a close with the development of phonolites, which according to current geological opinion (F. PRÄGER pers. comm.) cannot be younger than the brown coal of the Lausitz area, consequently the phonolites can be regarded with reasonable certainty as erupted between Upper Oligocene and high lower Miocene. In addition to the remnants of lava flows, dome

like intrusions with a characteristic columnar jointing are common. Dykes and necks are rare. Some autometamorphism is also found.

Thus according to the stratigraphical information available, the Lausitz volcanics belong in the greater part, to the first phase of activity, as defined by KOPECKÝ (1966), although some would place the activity in the second phase.

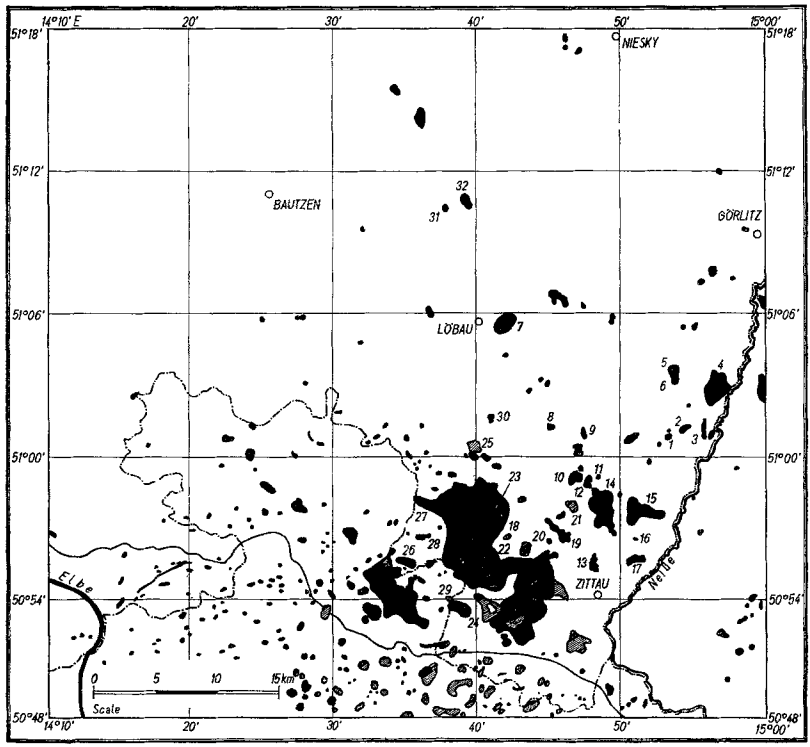


Fig. 1 b. Map of the Lausitz area (number 2 on figure 1 a) with the sites visited numbered to correspond to the numbers given in table 1.

Palaeomagnetic investigation

In the field six or seven oriented blocks were collected at each outcrop. Sampling was spread over the available outcrop consequently sample spacing may vary from a spread over 200 m to a grouping over 20 to 30 m. From each block a single core was cut for palaeomagnetic work and the adjacent core for petrological examination.

a) Direction of Magnetization. The direction of natural remanent magnetization (NRM) was measured under a short period astatic magnetometer. All samples were then progressively demagnetized in alternating magnetic fields increased in four steps up to 340 oe. The direction and intensity of magnetization was measured after each stage

in the demagnetization procedure, and using an electronic computer programme, the corresponding site mean values were computed. This has become a standard technique in the investigation of volcanic rocks. It was modified somewhat for those rocks too weakly magnetized to be

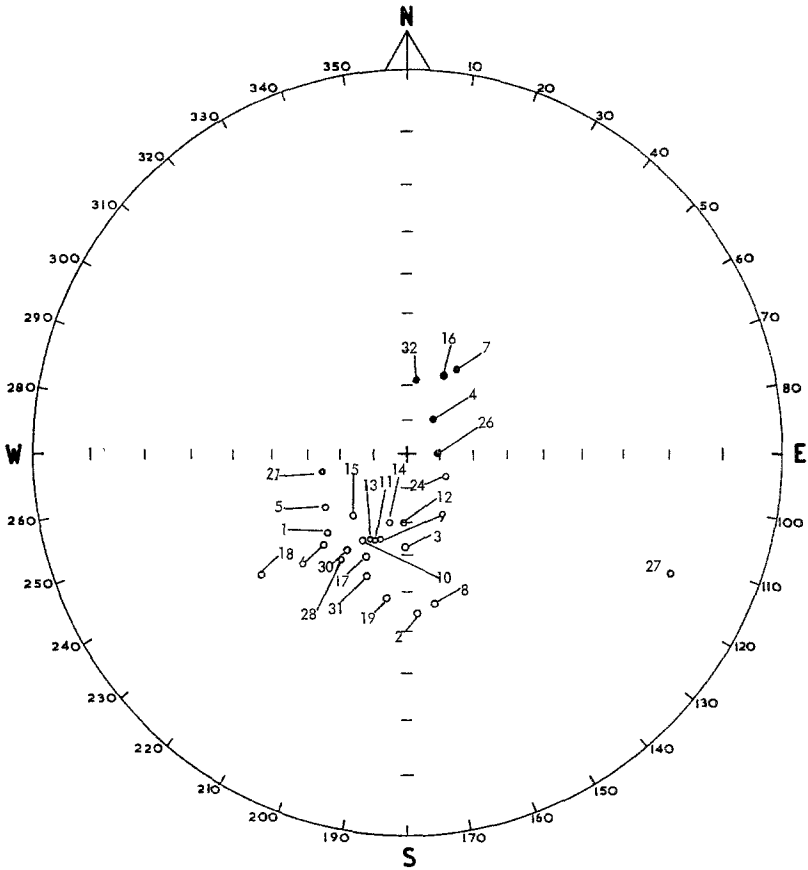


Fig. 2. Site mean directions of magnetization. The standard convention of open circles for projection on the upper hemisphere (reversed magnetization) and solids points for projection on the lower hemisphere (normal magnetization) is followed. The numbers refer to the individual sites listed in table 1.

measured under the low sensitivity magnetometer. The rocks in question, largely phonolites, were only demagnetized at the lower stages (85 and 170 oe) and measured under a high sensitivity long period magnetometer and means calculated as before.

The best representation of the original direction, that is the direction believed to record the direction of the magnetic field at the collecting site at the time the rock cooled, subsequent parasitic magnetizations

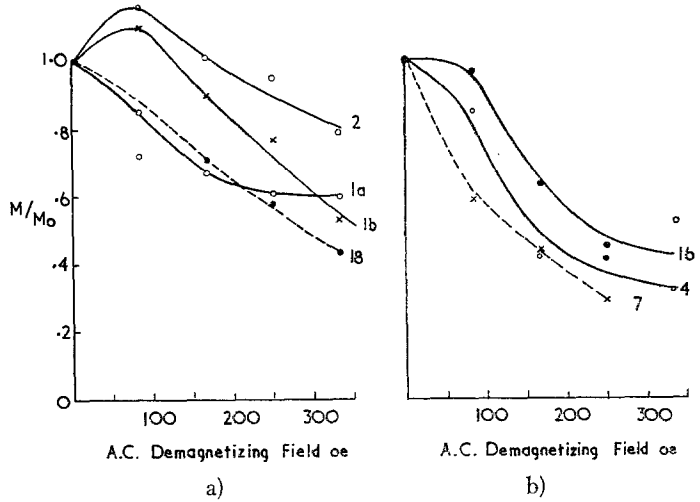


Fig. 3. Normalized alternating field demagnetization curves from a) reversely magnetized samples, b) normally magnetized samples. Numbers refer to the sites as in table 1.

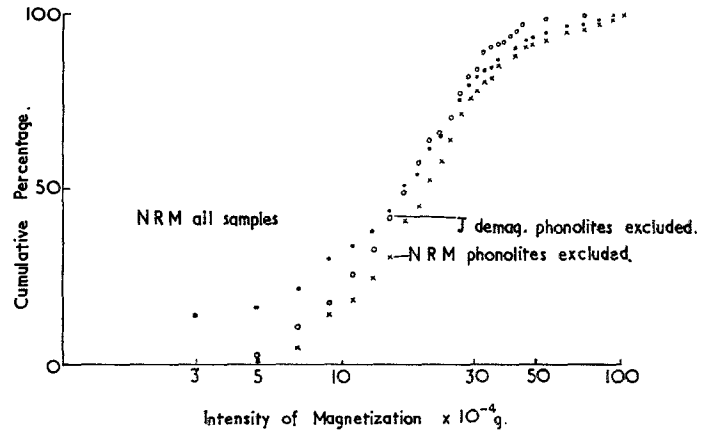


Fig. 4. Distribution of intensity of magnetization in the basalt samples shown as a cumulative percentage curve. The intensity scale is logarithmic. ● NRM all samples; o intensity after demagnetization, phonolites excluded; x NRM phonolites excluded.

removed, is regarded as that at which the circular standard deviation is a minimum. This is considered to be a better measure than k (FISHER 1953) when the sample size is small ($N < 8$). In cases where this figure coincided with the measurement of NRM the condition was imposed that subsequent demagnetization should not significantly change the mean direction. Arbitrarily, values of circular standard deviation in excess of

Table 1

Site No.	Site and/or rock type	Magnetization		Sampling a/b **	Confidence α°	C.S.D. ***	C.S.E. ***
		Declination	Inclination				
1	*(L-11) Steinberg bei Ostritz nepheline basanite	224.9	— 56.9	7/7	3.1	4.2	1.6
2	(L-19) Alter Hütberg bei Ostritz feldspar basalt	175.9	— 44.1	6/6	9.5	11.4	4.6
3	(L-10) Hütberg bei Ostritz nepheline basalt	181.6	— 62.5	6/6	6.4	7.7	3.2
4	(L-27) Steinbruch bei Leüba tephrite basalt	36.8	+ 77.6	6/6	2.9	3.5	1.4
5	(L-9) Hütberg bei Schönau nepheline basalt	236.8	— 60.4	6/7	7.4	8.9	3.6
6	(L-51) Kleine Hütberg bei Schönau nepheline basalt	222.9	— 53.6	6/6	10.5	12.6	5.2
7	(L-1*) Löbauerberg nepheline basalt	30.4	+ 61.0	5/7	11.3	11.8	5.3
8	(L-44) Hütberg bei Hernhut nepheline basalt	168.6	— 46.4	6/6	5.3	6.4	2.6
9	(L-58) Eichlerberg bei Rennersdorf feldspar basalt	200.1	— 62.1	6/6	5.5	6.6	2.7
10	(L-12) Eisberg bei Großhennersdorf	207.9	— 61.1	6/6	5.4	6.5	2.6
11	(L-13) Windmühlenberg bei Großhennersdorf nepheline basanite	198.3	— 63.4	6/6	3.8	4.5	1.9
12	(L-36) Pfältzens-Steinbruch bei Großhennersdorf feldspar basalt	188.3	— 69.9	7/7	4.4	5.9	2.2
13	(L-14) Eckhardtsberg bei Zittau nepheline basanite	201.5	— 61.9	7/7	4.2	5.6	2.1
14	(L-38) Schanzberg bei Oberseifersdorf nepheline basanite	193.4	— 69.2	7/7	4.3	5.7	2.2
15	(L-46) Steinbruch bei Wittgendorf nepheline basanite	220.7	— 66.6	7/7	6.8	8.2	3.3
16	(L-49) Buchberg bei Wittgendorf feldspar basalt	26.3	+ 64.7	6/6	5.7	6.9	2.8

* Numbers given in parentheses refer to site numbers in VOLLSTÄDT 1965.

Vector R	Demagnetizing Field oe	Log. Mean Intensity of Magnetization unit 10^{-4} e. m. u./3 cm + log Standard deviation (S.D.)				Log. Mean Susceptibility unit 10^{-4} e.m.u./3 cm + log Standard deviat.		Site No.
		NRM	S.D.	Demag.	S.D.	Suscept.	S.D.	
6.984	85	1.08	0.26	1.08	0.18	1.25	0.23	1
5.902	170	1.17	0.25	1.27	0.25	1.29	0.14	2
5.955	85	1.33	0.13	1.36	0.12	0.99	0.22	3
5.991	0	1.63	0.14	1.63	0.14	1.17	0.06	4
5.939	170	1.15	0.13	1.00	0.12	1.38	0.04	5
5.879	170	1.00	0.12	1.05	0.09	1.38	0.06	6
4.915	85	1.37	0.06	0.88	0.08	1.56	0.09	7
5.969	85	0.99	0.14	1.02	0.13	1.09	0.29	8
5.967	170	1.36	0.11	1.34	0.18	1.06	0.09	9
5.968	85	1.60	0.15	1.58	0.16	0.86	0.16	10
5.984	170	1.41	0.12	1.32	0.11	1.17	0.11	11
6.968	85	1.58	0.13	1.54	0.09	1.34	0.16	12
6.971	85	1.32	0.13	1.33	0.10	0.99	0.27	13
6.970	85	1.31	0.29	1.23	0.25	0.80	0.29	14
6.949	0	1.25	0.18	1.25	0.18	0.76	0.34	15
5.951	85	1.94	0.07	1.63	0.17	0.91	0.07	16

** Sampling b = samples collected; a = samples used in calculations.
*** C.S.D. = circular standard deviation; C.S.E. = circular standard error.

Table 1

Site No.	Site and/or rock type	Magnetization		Sampling a/b **	Confidence α°	C.S.D./C.S.E. ***	
		Declination	Inclination				
17	(L-40) Steinberg bei Hirschfelde nepheline basanite	201.9	— 57.2	6/6	5.9	7.1	2.9
18	(L-17) Koitsche bei Zittau phonolite	231.1	— 36.6	6/6	4.7	5.6	2.3
19	(L-16) Scheibenberg bei Hörnitz nepheline tephrite	188.2	— 48.3	7/7	6.9	9.2	3.5
20	(L-15) Schülerberg b. Zittau phonolite	No result		0/6	—	—	—
21	(L-39) Geiersberg bei Niederoderwitz phonolite	258.9	— 63.9	6/7	14.8	17.5	7.2
22	(L-20) Steinberg bei Hainewalde phonolite	No result		0/6	—	—	—
23	(L-37) Steinbruch bei Spitzkunnersdorf phonolite	No result		0/7	—	—	—
24	(L-22) Hütberg bei Hainewalde phonolite	117.6	— 76.3	6/7	7.3	8.8	3.6
25	(L-21) Kottmar bei Walddorf phonolite	No result		0/7	—	—	—
26	(L-44) Windmühlenberg bei Seifhennersdorf phonolite	90.2	+ 80.8	4/7	none	28.9	14.4
27	(L-41) Finkenberg bei Seifhennersdorf phonolite	113.9	— 15	7/7	none	57.9	21.9
28	(L-45) Großer Stein bei Seifhennersdorf phonolite	212.0	— 53.3	6/6	9.9	11.9	4.9
29	(L-42) Richterberg bei Neuspitzkunnersdorf phonolite	No result		0/6	—	—	—
30	(L-43) Steinbruch bei Neuoberruppersdorf nepheline basanite	212.7	— 56.4	6/6	11.4	13.6	5.6
31	(L-5) Strohmberg bei Weissenberg basalt	198.5	— 52.9	7/7	5.3	7.1	2.7
32	(L-6) Schafsberg bei Baruth basalt	6.7	+ 68.2	7/7	7.6	10.1	3.8

Vector R	Demagnetizing Field oe	Log. Mean Intensity of Magnetization unit 10^{-4} e. m. u./3 cm + log Standard deviation (S.D.)				Log. Mean Susceptibility unit 10^{-4} e.m.u./3 cm + log Standard deviat.		Site No.
		NRM	S.D.	Demag.	S.D.	Suscept.	S.D.	
5.961	170	1.29	0.21	1.23	0.19	1.09	0.23	17
5.976	85	0.77	0.10	0.75	0.09	0.35	0.06	18
6.923	85	1.17	0.14	1.24	0.06	1.51	0.04	19
—	—	—	—	—	—	1.23	0.51	20
5.767	170	—	—	—	—	1.01	0.04	21
—	—	—	—	—	—	0.92	0.07	22
—	—	—	—	—	—	0.87	0.02	23
5.942	0	0.29	0.17	0.29	0.17	1.13	0.03	24
—	—	—	—	—	—	0.42	0.28	25
3.642	0	0.57	0.40	0.57	0.40	1.19	0.05	26
4.157	0	$\bar{1}$.85	$\bar{1}$.94	0.85	$\bar{1}$.94	0.37	0.14	27
5.925	0	$\bar{1}$.96	0.16	$\bar{1}$.96	0.16	0.31	0.11	28
—	—	—	—	—	—	1.08	0.05	29
5.869	85	1.17	0.27	1.24	0.21	0.14	$\bar{1}$.76	30
6.955	85	1.43	0.09	1.45	0.10	1.24	0.16	31
6.907	0	1.33	0.11	1.33	0.11	1.46	0.02	32

20 were not considered significant and are so indicated in table 1 which summarizes the statistical data. It can be seen in the table that in the majority of cases, the circular standard deviation was below half his figure. Of the phonolites measured under the high sensitivity magnetometer the directions found were scatted at all sites (sites 20, 22, 23, 29) save one (site 21). The site mean directions tabulated are shown graphically in figure 2, with site numbers conforming to those in the table.

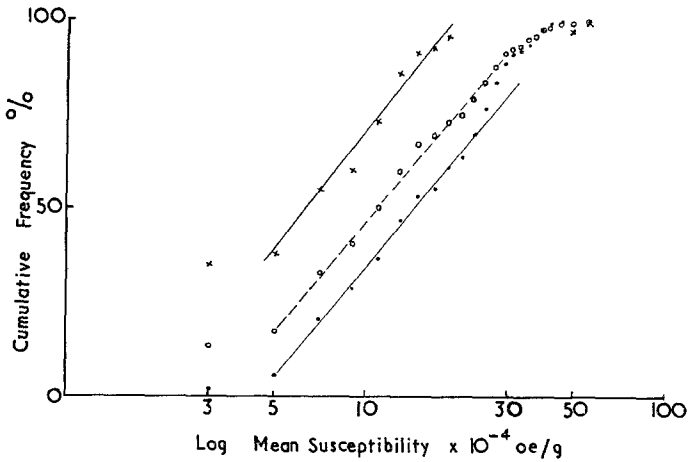


Fig. 5. Distribution of susceptibility plotted as a cumulative percentage curve. The susceptibility scale is logarithmic. Straight lines are drawn as guides only: x phonolites, ● basalts, o combined.

b) Intensity of magnetization. Intensity of magnetization was measured concurrently with measurements of the direction of remanence. In table 1 the site mean intensities of magnetization and standard deviation are given in logarithmic form. Values are given for NRM and for the demagnetization stage of minimum circular standard deviation. A number of normalized demagnetization curves from samples showing both normal and reversed magnetization are shown in figure 3. The distribution of the intensity of magnetization within the suite of basalt samples, both of NRM and demagnetized, is given in figure 4. In this figure the cumulative percentage frequency of intensity values is plotted against intensity whose scale is logarithmic.

c) Susceptibility. Volume susceptibility was measured on a transformer bridge (Collinson, Molyneaux, Stone 1963). No measurements of anisotropy of susceptibility were attempted. Mean site values were computed and given in logarithmic form in table 1. The cumulative percentage frequency curve, figure 5, in which curves are given for the distribution with and without the inclusion of the phonolites, may be compared with the intensity distribution in figure 4.

Petrological investigations

The basaltic rocks of Lausitz can be divided petrographically into three closely related types, the nepheline basanites, the nepheline basalts and the plagioclase basalts. The principal melanocratic minerals are augite, olivine and magnetite, the leucocratic components are nepheline and glass. There are, in addition late crystallizing zeolites and calcite. Olivine also appears as phenocrysts, commonly with corroded margins. In part the phenocrysts may be altered to iddingsite, not infrequently with picotite inclusions. A detailed description of the Lausitz basalts has been given by Schreiter (1965).

There are several varietes of the greenish gray to gray coloured phonolites. The most important are the trachytoid and nephelinoid types, a third and rarer form is a tephritic phonolite. The principal minerals are sanidine, anorthoclase, nepheline and sodic-pyroxene, the secondary minerals include hornblende, apatite, sphene, and titanomagnetite.

As a result of research in recent years by Ade-Hall (1964), Ade-Hall et al. (1965), Watkins & Haggerty (1965), Vollstädt (1965), and Wilson (1964), the detailed study of the ore minerals, with particular reference to their oxidation state (Buddington & Lindsley 1964) has become relevent to the question of reversed magnetization. Particularly is this true when attempts are made to use the sign of magnetization stratigraphically (Roche 1953, Rutten 1964, Nairn 1962, 1966). Although the problems of precise stratigraphy in the Lausitz area render attempts at a magnetic correlation highly tentative, the petrography of the ore minerals was studied to see whether there could be found any correlation between the oxidation state and the sign of magnetization.

The principal ore-mineral present, and also the most important from the paleomagnetic viewpoint was titano-magnetite. The description of the titano-magnetite however is better based on size and shape rather than degree of oxidation (see Watkins and Haggerty 1965, Ade-Hall, Wilson and Smith 1965). There are on the one hand the predominant rounded to idiomorphic titanomagnetite grains and on the other highly irregular grains. The relationship between these forms and palaeomagnetic stability has already been described (Vollstädt 1965).

The average grain size was 10 μ although grains in the range 1—5 μ were not uncommon (e. g. site 16 Buchberg bei Wittgendorf). At the opposite extreme at the Löbauerberg outcrop (site 7) an average grain size of 50 μ was recorded. The magnetic ore phase content varied from less than 1% (e. g. the phonolite of Kottmar bei Waldorf site 25, and site 18 Koitsche bei Zittau) to greater than 10% (e. g. site 31 Stromberg bei Weissenberg and site 32 Schafsberg bei Baruth).

Optical examination with oil immersion indicates that in most cases the ore mineral is a homogeneous titanomagnetite which would be placed in Class II up to Class III of the Watkins/Wilson grouping based on degree of oxidation. Mixed titanomagnetites are comparatively seldom found. Samples from site 5, Hutberg bei Schönau, contain two distinct

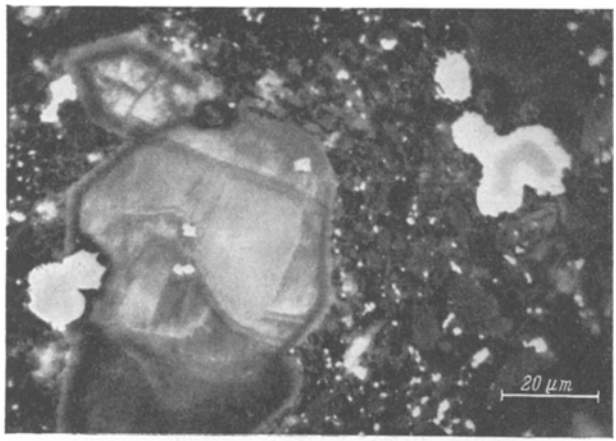


Fig. 6. Titanomagnetite in a nepheline basanite from Scheibenberg, Hornitz.

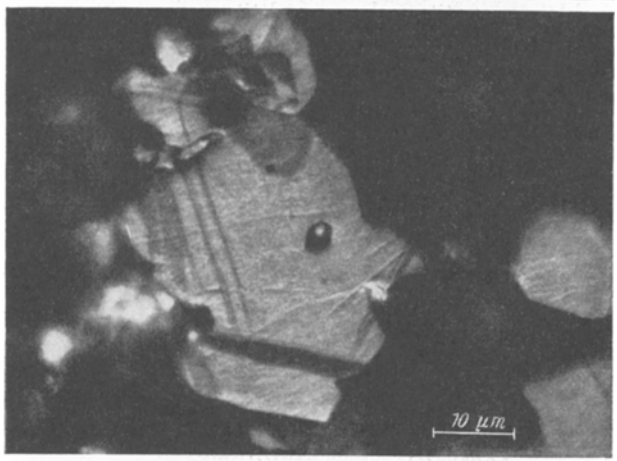


Fig. 7. Titanomagnetite in a feldspar basalt from Eichlerberg, Rennersdorf.

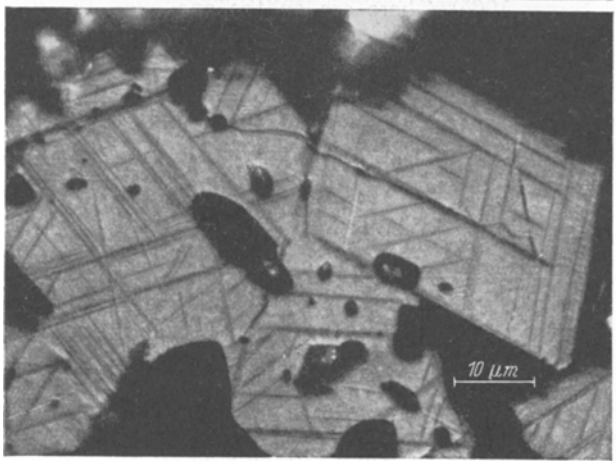


Fig. 8. Titanomagnetite from Hütberg, Schönau showing lamellae.

titanomagnetite forms, fig. 6 illustrates a typical exsolution pattern with lamellae making an angle of 60°, while fig. 7 illustrates the second type in which strong ilmenite lamellae appear.

The titanomagnetites from some outcrops (e. g. site 7 Löbauer Berg) shows marginally, a higher reflectivity suggesting a migration of titanum into these regions. This supposition is strengthened by the occurence of a Ti-augite in rocks from this outcrop.

The remaining magnetic minerals in a descending order of importance are picotite, maghemite, and haematite. Picotite is always associated with olivine and only occurs in olivine bearing rocks (fig. 8). Maghemite although difficult to identify has been found in most basalts (see also SCHREITER 1965). Haematite and pyrite occur only in insignificant amounts.

In some phonolites secondary magnetite occurs as a hornblende alteration product.

Discussion of results

a) Directions of Magnetization. The indeterminacy surrounding the exact age of the Lausitz volcanics precludes any certainty in relating the sign of magnetization to a particular geological age. Bearing in mind the limitation that certain of the basalts could conceivably be younger than the phonolites the following magnetic sign sequences are possible.

	1	2	3	4
phonolite	+	—	+	—
	—	+	—	+
basalt	+	—	—	+
	—	+	+	—

Inspection of table 1 indicates that all but five of the sites are reversely magnetized, from which we tentatively conclude that model 3 provides the best working basis. Models 1 and 2 are excluded because they require three changes of sign instead of the two required by 3 and 4. Model 3 is preferred over 4 because it places all the reversals in a single zone, and is therefore, the simplest solution. Obviously, there can be no guarantee that it is necessarily the correct one (further research by one of the authors H. V. is directed to this problem).

Although there are a number of palaeomagnetic studies of other European volcanics rocks close to the Oligocene-Miocene boundary, the stratigraphical control both locally and regionally is in general rather poor. In consequence the palaeomagnetic zonal sequence cannot be regarded as firmly established. The following sequence (table 2) summarizes the pioneering work of ROCHE (1953) which was reviewed by RUTTEN (1960) and the results of some preliminary work in Germany (NAIRN, 1962). It is possible to fit, tentatively, the Lausitz volcanics into this scheme and so suggest a middle upper Chattian to lower Aquitanian age for most of the volcanic activity. This is certainly consistent with the Chattian age for

the Lausitz tuffs. It would mean that the Lausitz activity was coeval with the basalts of the Westerwald (NAIRN, 1962).

Table 2. Tentative palaeomagnetic zones of the Upper Oligocene and Lower Miocene.

L—M Miocene	Vindobonian	Tortonian		
		Helvetian		
	Burdigalian			
Oligocene	Aquitanian	Chattian	U N	22.8 m. y.*)
			M R	
	Stampian		L N	Lausitz volcanics
		Rupelian		
	Sannoisian		R	32 m. y.**)

*) 22.8 m. y. K/A age of the Drachenfels trachyte, the later Wolkenberg latite is normally magnetized.

**) 32 m. y. He age of a Polish basalt belonging to the N. Bohemian province. The sign of magnetization is known (unpublished data).

b) Susceptibility and intensity of Magnetization. The distribution of susceptibility and intensity of magnetization through the collected suite of samples is shown in figs. 4 and 5. Each sample has been given unit weight as the dispersion within a site is great as the differences between sites. Both intensity and susceptibility curves have a similar form and both follow closely a log normal distribution. Apart from a few percent in which high suceptibility and intensity values are found, no inflections are found in the curves. This is not so in certain andesite suites (see NAIRN 1967, in press). It may thus be evidence that the Lausitz volcanics belong to a single suite.

c) Petrology. The petrological and ore microscopical examination has a bearing on the question of magnetic stability as well as on the mineralogical self reversal problem. In the majority of samples a homogeneous titanomagnetite was found, exsolution lamellae of ilmenite and the oxidation alteration of the ore minerals are relatively uncommon. Yet frequently, stable and unstable samples are found in close proximity and both normal and reversed magnetized samples occur. Elsewhere (VOLLSTÄDT, 1967) a satisfactory experimental basis for these observations has been given.

Consideration of normally and reversely magnetized basalts and phonolite fails to disclose any difference in the magnetic minerals in the majority of cases investigated. The absence of more than one magnetic phase more or less excludes the possibility of a mineralogical self reversal. The remaining possibility, the differential occupation of A and B sites in the titanomagnetite lattice is excluded by the results of neutron diffraction studies (KIRSTEN et al., 1966), these have, however, been carried out at only one

locality (site 18) because of the great amount of time and material required, and they indicate here that the only possibility is of a geomagnetic field reversal. That is to say that during the eruption of the phonolites the magnetic field was reversed.

In the case of the basalts the position is more complex and detailed study similar to that of the phonolite above is in progress.

Conclusions

Although no definite conclusions can be reached about the exact age of the magnetization because of the geological uncertainties, the reversed magnetization is consistent with what is known in rocks of Middle to Upper Chattian to Lower Aquitanian age in other regions. The uniformity of the distribution of susceptibility and intensity of magnetization, both of which follow a log normal distribution may be an indication of the uniformity of the Lausitz volcanics and hence militate against the suggestion that some basalts may be younger than the phonolites. Further prediction is dangerous for the Lausitz volcanics form only a small part of the N. Bohemian volcanic province.

The petrology of the ore minerals is such as to rule out with certainty in at least one case a mineralogical origin for the reversed sign of magnetization. Whatever may be the case in other localities here at least the evidence points to a geomagnetic field reversal.

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Beobachtungen zur Stratigraphie des Schilfsandsteins im ost-lippischen Bergland

(Lithostratigraphische Untersuchungen in Sandstein-Folgen III)

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Mit 9 Abbildungen

Zusammenfassung

Die sandige Fazies des Schilfsandsteins ist in Ost-Lippe rhythmisch aufgebaut. Diese Abfolge ist so konstant, daß sie als stratigraphische Gliederung verwendbar ist. Einzelne Schichtglieder lassen sich bis maximal 2500 m Erstreckung verfolgen und als Leithorizonte benutzen. Das gilt auch beim Übergang in andere Faziesbereiche. Die Sandsteine selbst sind ebenfalls (z. T. undeutlich) rhythmisch entwickelt; die Zuordnung von Rhythmen ist in einigen Fällen möglich. Modalbestands-Analysen, Korngrößen- und Karbonat-Bestimmungen ergeben keine feinere Untergliederungsmöglichkeit der Schichtfolge. Sie vermögen die makroskopische Schichtgliederung teilweise zu stützen bzw. zu ergänzen. — Ausblick auf Sedimentationsverhältnisse im Untersuchungsgebiet und arbeitsmethodische Fragen.

Abstract

The sandstone-facies of the Schilfsandstein (km_2) in the eastern part of Lippe (eastern Westphalia) shows a rhythmic sedimentation, which may be used as a

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